SANJAY KUMAR YADAV

Civil Engineer | *Remote Sensing, GIS & Land-Use Change Modelling*

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RESEARCH PROFILE

Civil engineer with more than four years of experience in GIS and satellite remote sensing for land-use planning and disaster-risk assessment in Nepal. At Picasso Engineering Consultancy I reconstruct past land cover from historical Google Earth imagery and project future land-use change with Cellular Automata–Markov modelling in TerrSet, working mainly in ArcGIS Pro. I also process Sentinel-2 and Landsat imagery in Google Earth Engine and use Python to automate GIS workflows, and I classify land cover with Random Forest. I am seeking an M.S. to turn this applied experience into research on long-term landscape change, using historical imagery, reproducible workflows and machine learning.

RESEARCH INTERESTS

Long-term land-use and land-cover change; historical aerial and satellite imagery; vegetation and agricultural landscape dynamics; change detection and image classification with machine learning/GeoAI; land-change simulation and uncertainty; human–environment interactions; hazard and environmental risk assessment; integration of GIS and BIM (GeoBIM) and digital twins.

EDUCATION

B.E. in Civil Engineering | Pokhara University (Cosmos College), Nepal *Completed August 2023*

CGPA 3.09/4.00. Relevant coursework: Remote Sensing and GIS Applications in Civil Engineering; Surveying and Photogrammetry; Hydraulics and Hydrology; Construction Project Management.

RESEARCH AND PUBLICATIONS

Singh, S. K., Joshi, A. M., Bhattarai, P. K., Karki, S., **Yadav, S. K.**, Poudel, U., Sharma, N., Gouli, M. R., & Khadka, N. (2026). Integrating physical simulation and machine learning for flood susceptibility mapping in Kathmandu Valley. *Manuscript submitted to H2Open (IWA Publishing), ID H2Open-D-26-00293; under review.*

- My contribution: acquisition and preprocessing of geospatial datasets, GIS support for spatial analysis, and thematic maps of input data and results. Physical simulation and machine-learning modelling were carried out by other co-authors.

RESEARCH-RELEVANT PROFESSIONAL EXPERIENCE

Geospatial Analyst (part-time) | Picasso Engineering Consultancy *June 2024 – present*

- **Land-use change prediction:** for Dharan Sub-Metropolitan City I mapped land cover for 2004, 2016 and 2025 from historical imagery (Google Earth) and projected it to 2050 with Cellular Automata–Markov chain modelling in TerrSet; I applied the same approach to the Rakam Karnali new town (2015 baseline projected to 2055) and Chandannath Municipality (Jumla), supporting municipal and new-town planning.
- **GIS workflow:** ArcGIS Pro is the primary platform for about 80% of my GIS work (digitising, geoprocessing, overlay, suitability and map production), with TerrSet used for change modelling.
- **Python automation:** scripts inside ArcGIS Pro to automate geoprocessing, bulk raster downloads (Google Earth Engine) and report writing.
- **Planning assignments:** Risk-Sensitive Land Use Planning for Bharatpur Metropolitan City and Dharan Sub-Metropolitan City; GIS suitability and urban-development-potential analysis for the Bheriganga Valley new-town study (three municipalities of Surkhet District, for the DUDBC New Town Office) and for the Rakam Karnali small new town on the Madhya Pahadi Lokmarga; earlier UNDP-supported RSLUP for Chandannath Municipality, Integrated Land Development Planning for Guthichaur Rural Municipality, and a digital municipal profile for Phikkal Rural Municipality (KoboToolbox).
- **Remote sensing and hazard mapping:** Sentinel-2 and Landsat processing in Google Earth Engine (NDVI, NDWI); DEM-derived slope, TWI, flow accumulation and catchments; AHP-based multi-criteria hazard susceptibility mapping; Random Forest land-cover classification in ArcGIS Pro.

GIS & Structural Engineering Consultant | Arc Aid Planners & Developers Pvt. Ltd. *April 2024 – July 2025*

- Municipal land-use classification for Ajirkot, Tila and Sinja rural municipalities (Nepal).

GIS Analyst | Scholars Consult Pvt. Ltd. *Oct – Dec 2024*

- Boundary digitising, land-cover mapping and land-use zoning classification for Kalika Municipality, Chitwan.

GIS & Data Support Consultant | Practical Action Consulting (USAID-funded CVRA) *Aug – Sept 2024*

- Spatial data management, vulnerability-matrix calculation and risk mapping for a climate vulnerability and risk assessment.

Civil Engineer & GIS Engineer | Map Nepal Engineering Consultancy Pvt. Ltd. *July 2022 – May 2024*

- Municipal land-use classification and zoning for seven local governments using GIS and remote-sensing data; transport master-plan inputs, digital municipal profiles and road-corridor models.

OTHER PROFESSIONAL EXPERIENCE

BIM Engineer Coordinator (full-time) | AMY METOM Engineering Consultancy *January 2026 – present*

- Infrastructure BIM (since mid-2026): using Civil 3D, InfraWorks and Navisworks, I bring drone-survey terrain models (DTM) and imagery together with existing conditions, designed corridor alignments and pipe networks from other design software, and run clash detection in Navisworks. This work combines geospatial data with design models and is my practical route into GIS–BIM integration.
- First six months: coordinated multidisciplinary building BIM (structural, architectural, MEP) for international projects in Revit, including model review and quality control.

Earlier structural and construction experience: Shah Engineering Consultancy, freelance design work and Local Roads Bridge Support Unit internship (2022–2024).

TECHNICAL SKILLS

GIS and remote sensing: ArcGIS Pro (advanced); QGIS; Google Earth Engine; Google Earth historical imagery; Sentinel-2, Landsat 8/9, ALOS PALSAR; land-cover classification; NDVI/NDWI; overlay, suitability and AHP analysis.

Land-change modelling: TerrSet; Cellular Automata–Markov chain simulation of future land use.

Programming and GeoAI: Python scripting inside ArcGIS Pro for geoprocessing automation; automated raster downloads (Google Earth Engine) and report-generation scripts; Random Forest land-cover classification (ArcGIS Pro); GeoPandas, Rasterio, GDAL, NumPy, Pandas, scikit-learn.

Terrain, UAV and field data: DEM, slope, TWI, flow accumulation; DJI Mavic 3E, Pix4D, orthomosaics, DSM/DEM; KoboToolbox; GPS data collection.

Engineering: Civil 3D, InfraWorks, Navisworks, Revit, AutoCAD, ETABS, SAP2000.

TRAINING AND REGISTRATION

- Remote Sensing and GIS Application, National Innovation Centre, Nepal (2022); Application of ArcGIS in Civil Engineering, Cosmos College (2021); Climate-Resilient Agriculture irrigation design, ICIMOD (2023).
- Registered Civil Engineer, Nepal Engineering Council, No. 84549 (2024).
- Languages: English (advanced professional; language of instruction through B.E.), Nepali (fluent), Hindi (professional), Maithili (native).

REFERENCES

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